

AN AUTOMATED INTRADAY STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE

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ABSTRACT. We specify the routine that calibrates the five Heston (1993) parameters $\Phi = (\theta, \kappa, \eta, \rho, v_0)$ to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot S_{ref} , the moneyness-normalised and grid-snapped strike K^* , the surface-coverage requirements, the in-engine relative-price objective, the implied-volatility (“IV-space”) selection and acceptance gate, and the boundary-pegging rejection rule.

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1. The model

Under the risk-neutral measure, with constant risk-free rate r and continuous dividend yield g , the log-price $X_t = \ln S_t$ and instantaneous variance v_t of the SPX follow the Heston (1993) dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where W, B are independent Brownian motions, so $\text{corr}(dX_t, dv_t) = \rho dt$. The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with $\theta > 0$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\eta > 0$ the volatility of variance (QuantLib's **sigma**), $\rho \in [-1, 1]$ the leverage correlation, and $v_0 > 0$ the initial variance. The Feller quantity (Feller, 1951)

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit; $\mathcal{F}(\Phi) \geq 0$ guarantees $v_t > 0$ almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

1.1. Pricing operators. Fix a valuation with maturity $\tau > 0$ (in years), strike K , and type $w \in \{+1, -1\}$ (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (5)$$

Black price (Black, 1976; Black and Scholes, 1973). With total standard deviation $\Sigma = \sigma\sqrt{\tau}$,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (6)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where $\mathcal{N}$ is the standard normal CDF and $F = F(S, \tau)$.

Heston price. The Heston model price $C_H(\Phi; S, K, \tau, w)$ is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (7)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for $j \in \{1, 2\}$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (8)$$

and characteristic functions $f_j = \exp(C_j + D_j v_0 + iu \ln F)$ given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[(b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (9)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (10)$$

where $\mu_1 = \frac{1}{2}$, $\mu_2 = -\frac{1}{2}$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$. In code this integral is evaluated by the QuantLib (QuantLib) `AnalyticHestonEngine` (which uses a branch-stable formulation of (8)); equations (7)–(8) define the operator it realises.

Implied volatility. The Black-implied volatility of a price C is the inverse of (6) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (11)$$

computed numerically (QuantLib `blackFormulaImpliedStdDev` on the forward (5), returning Σ , then $\sigma = \Sigma/\sqrt{\tau}$). This operator defines the IV-space residuals of §5.

2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day d , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by i , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (12)$$

being respectively the quote timestamp, strike, type $w_i \in \{+1, -1\}$, traded size, trade price, trade-implied volatility, the underlying level *at trade time* t_i , and the days to maturity. Maturities enter the engine as τ_i in years under Actual/365, i.e. $T_i = \tau_i/365$.

The day is rate-stamped with $r = r(d)$ and $g = g(d)$, the most recent risk-free and dividend rates on or before d (a flat-forward curve for each; under flat curves the evaluation date is immaterial to T_i , so it is set to the QuantLib “today”).

Pre-filter. We keep only economically usable, Heston-appropriate quotes:

$$\mathcal{D} = \{ i : \sigma_i > 0 \text{ and } \tau_i \geq \tau_{\min} \}. \quad (13)$$

The floor $\tau_{\min}$ (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive η, κ to Feller-violating extremes that pollute the pooled fit.

3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot S , yet the trades in $\mathcal{D}$ occur across an intraday range of underlying levels $\{S_i\}$. We reconcile the two with one reference spot and a moneyness re-striking.

Reference spot. Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (14)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (15)$$

A flagged day ($R > R_{\max}$) is still calibrated and written, but the normalisation below is recorded as strained (its S_{ref} should be treated with suspicion).

Moneyness normalisation and grid snap. Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the

SPX strike grid of width Δ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (16)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in K/S rather than in the absolute strike. Re-striking by (16) therefore re-centres the whole day onto S_{ref} so that trades executed at different S_i but the *same* moneyness share a common surface column. The grid snap collapses the continuum of K_i^* values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract's *original* (S_i, K_i) , since the calibrated Φ is spot-independent.

4. Surface construction

From the normalised trades we build a partial $K^* \times \tau$ implied-volatility surface, selecting the most liquid, nearest-the-money quotes.

Maturity selection. For each distinct maturity τ present in $\mathcal{D}$, let $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$ be its traded volume. Keep the N_t highest-volume maturities,

$$\mathcal{T} = \text{top-}N_t \text{ maturities by } V(\tau). \quad (17)$$

Strike selection (per maturity, per wing). Within each kept maturity $\tau \in \mathcal{T}$, separate calls ($w_i = +1$, OTM with $K_i^* \gtrsim S_{\text{ref}}$) from puts ($w_i = -1$, OTM with $K_i^* \lesssim S_{\text{ref}}$) and keep the n_k nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{n_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{n_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (18)$$

Pivot to a surface. For each retained cell (k, τ) with $k \in \mathcal{K}_\tau$, set the surface entry to the implied volatility of the *latest* trade in that cell,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (19)$$

and leave it missing otherwise. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (20)$$

be the set of populated cells; these become the calibration targets (helpers).

Coverage requirement. To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (21)$$

A day failing (21) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

5. The optimization problem

Let $\mathcal{N}$ index the targets with market data $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$, all priced at the single spot S_{ref} . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^5 [\ell_j, h_j] \subset \mathbb{R}^5, \quad (\ell, h) \text{ in the } (\theta, \kappa, \eta, \rho, v_0) \text{ order of Table 2.} \quad (22)$$

5.1. Residuals. Two residual systems are used: a price-space system that the local optimizer minimises, and an IV-space system that selects among restarts and gates acceptance. The in-engine objective is configurable. It defaults to the price-space system below, and can instead be set to the IV-space system of (25). The choice changes only what each restart minimises. Selection (30) and the acceptance gate (32) always run on the IV-space error J_{iv} , so they are unaffected by it.

Price-space (in-engine objective). For each target the market value is the Black price at the quoted vol, $C_n^{\text{mkt}} = C_B(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$, and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Phi) = \frac{C_H(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (23)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Phi) = \left(\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Phi)^2 \right)^{1/2}. \quad (24)$$

The per-target residual $\varepsilon_n^{\text{px}}$ is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt (Marquardt, 1963) is a sum-of-squares method, so it drives the squared residual norm $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Phi)^2$ to a local minimum (equivalently J_{px} , with which it shares a minimiser).

IV-space (selection and gate). Invert each Heston price back to a Black vol via (11) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Phi) &= \mathcal{I}(C_H(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Phi) &= \sigma_n^{\text{mod}}(\Phi) - \hat{\sigma}_n, \end{aligned} \quad (25)$$

with objective

$$J_{iv}(\Phi) = \left(\frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Phi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Phi) \text{ inverts finitely}\}. \quad (26)$$

J_{iv} is measured in volatility points; it is the natural surface-fit metric and, unlike J_{px} , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

5.2. Multi-start local optimization. A single start into a non-convex problem is fragile (Mikhailov and Nögel, 2003), so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right), \quad (27)$$

and define the seed set $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$ of $(v_0, \kappa, \theta, \eta, \rho)$ tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (28)$$

For each seed $s \in \mathcal{S}$ we solve the box-constrained nonlinear least-squares problem

$$\hat{\Phi}_s = \arg \min_{\Phi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Phi)^2 \quad (\text{LM, started at } \Phi^{(0)} = s), \quad (29)$$

using Levenberg–Marquardt under the box constraint $\mathcal{B}$ (`QuantLib NonhomogeneousBoundaryConstraint`). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Phi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Phi}_s). \quad (30)$$

Ranking by J_{iv} (not J_{px}) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

6. Acceptance gate

Φ^* is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction δ , be

$$\text{PEGGED}(\Phi) = \bigvee_{j=1}^5 \left[(\Phi_j - \ell_j) < \delta (h_j - \ell_j) \quad \vee \quad (h_j - \Phi_j) < \delta (h_j - \ell_j) \right]. \quad (31)$$

The day is accepted iff

$$\boxed{J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Phi^*)}. \quad (32)$$

The first clause is the IV-space gate ($\varepsilon_{\text{iv}} \approx 2$ vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of J_{iv} . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

Why an IV-space gate. An earlier version gated on J_{px} directly. On a pooled day rich in deep-OTM wings, J_{px} is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with J_{iv} aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is ≈ 0.8 – 1.4 vol points, and the residual rejections are boundary-pegged κ or ρ (a separate identification lever), not fit-quality failures.

7. Output and diagnostics

An accepted day yields one parameter row keyed by date,

$$(d, S_{\text{ref}}, r, g, \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*, \mathcal{F}(\Phi^*), J_{\text{iv}}, J_{\text{px}}, \dots, R, \text{HIGH_MOVE}), \quad (33)$$

together with coverage counts ($|\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|$) and total volume. Because the calibration is once-daily, these rows accumulate into a single file, regenerated each run from the accepted days. For repricing diagnostics each retained contract is valued under Φ^* at its *original* (S_i, K_i) , not (S_{ref}, K^*) , since the structural parameters are spot-independent; this gives an honest comparison against the actual trade price P_i .

8. Constants

Symbol	Code name	Value	Role
$\tau_{\min}$	MIN_DTM	7 days	minimum maturity (drop ultra-short)
$R_{\max}$	MAX_MOVE_PCT	0.03	intraday-range flag threshold
Δ	STRIKE_GRID	5.0	SPX strike lattice for K^* snap
N_t	MAX_NT	12	maturities kept (by volume)
n_k	MAX_NK	8	strikes kept per wing
M_t	MIN_MATS	3	minimum maturities in surface
M_k	MIN_STRIKES	5	minimum strikes in surface
M_{cell}	MIN_CELLS	12	minimum populated cells $ \mathcal{N} $
ε_{iv}	IV_RMSE_ACCEPT	0.02	IV-space acceptance gate (vol pts)
δ	BOUND_TOL	10^{-3}	boundary-pegging tolerance fraction

TABLE 1. Surface, selection, and gate constants.

	θ	κ	η	ρ	v_0
lower ℓ	10^{-4}	10^{-2}	10^{-2}	-0.999	10^{-4}
upper h	1.0	20.0	2.0	0.5	1.0

TABLE 2. Box bounds $\mathcal{B}$, in the `model.params()` order $(\theta, \kappa, \eta, \rho, v_0)$. The ρ upper bound is kept slightly positive (equity leverage implies $\rho < 0$) and η is capped at 2.0 (SPX vol-of-vol ~ 0.3 – 1.2).

9. Empirical findings

We ran the routine over the full CBOE SPX trade history available to us, spanning 3,215 trading days from 3 January 2012 to 9 October 2024. Unless noted, the figures below are for the default price-space objective (24); the IV-space objective (26) is compared at the end of the section. Of these days, 1,713 (53.3%) passed the acceptance gate (32) and contribute a parameter row; the 1,502 rejected days are logged with their cause. The rejection split is dominated by boundary pegging: 1,398 days (93% of rejections) are rejected for a pegged parameter, 103 for an IV-space miss ($J_{\text{iv}} > \varepsilon_{\text{iv}}$), and 1 for having no usable trades. Pegging (almost always $\kappa \rightarrow 20$ or $\rho \rightarrow -0.999$ hitting the box wall) is therefore the binding constraint on the accept rate, not surface quality.

Fit quality. On the accepted set the surface fit is excellent. The IV-space error (26) has a median of 0.48 vol points (mean 0.55, 90th percentile 0.89, maximum 1.93), so J_{iv} sits far inside the 2-vol-point gate on essentially every accepted day and never binds in practice. Acceptance is governed by the no-pegging clause alone. An independent repricing check (each contract valued under Φ^* at its *original* (S_i, K_i) and compared to the trade price) confirms this: the per-day median relative price error is 6.7% and its 90th percentile 10.6%, figures inflated by the deep-OTM wings whose near-zero prices make any absolute miss a large relative one. This is exactly the effect that motivated gating in IV space rather than price space (§6). A separate validator clears 1,708 of 1,713 accepted days (99.7%); the 5 hard failures are repricing-fit outliers, not parameter pathologies.

Parameter levels. The accepted parameters are economically sensible and stable across the twelve-year sample. The leverage correlation is consistently and

strongly negative, median $\rho = -0.78$ (10th–90th percentile $[-0.89, -0.62]$), as equity-index smiles require. Long-run variance median $\theta = 0.053$ ($\approx 23\%$ annualised vol) and initial variance median $v_0 = 0.016$ ($\approx 12.5\%$) bracket realistic SPX levels, with v_0 the lower of the two on the typical low-realised-vol day. Mean reversion median $\kappa = 2.19$ and vol-of-vol median $\eta = 0.89$ are likewise in the usual SPX range. Because the gate rejects any boundary-pegged fit, the accepted set is pegging-free by construction: only 14 days carry ρ below -0.99 and none reach the κ ceiling. A clean interior set is what survives the gate, not evidence that pegging is solved on the underlying population.

Feller condition remains a standout caveat. The gate does not test Feller, and short-tenor SPX fits violate it routinely: $\mathcal{F}(\Phi^*) = 2\kappa\theta - \eta^2 < 0$ on 1,701 of 1,713 accepted days (99.3%), with median $\mathcal{F} = -0.54$. The driver of the violation is a large vol-of-vol relative to $\kappa\theta$; η exceeds 1.5 on 146 days (8.5%) and approaches its 2.0 cap (max 1.99). A pinned v_0 that disagrees with the at-the-money level appears on only 4 days and a degenerate κ on 8, so among the validator’s soft flags Feller violation and an elevated η are by far the most common. Together they mark these fits as ones whose variance path can reach zero, even where the surface fit itself is tight.

Price-space versus IV-space objective. The in-engine objective is configurable (§5), and every figure above uses the price-space default. Re-running the full sample with the IV-space objective (26) tests whether minimising in the gate’s own units helps. It does not. The IV-space objective accepts 1,645 days (51.2%), which is 68 fewer than the price objective, and its rejections are even more dominated by pegging (1,479 of 1,570, or 94%). On the accepted days the fit in its own metric is marginally tighter, with a median J_{iv} of 0.45 vol points against 0.48 and a 90th percentile of 0.83 against 0.89. The parameter levels move within sampling noise (median $\rho = -0.78$, median $\theta = 0.054$, median $\kappa = 2.12$, median $\eta = 0.90$, median $v_0 = 0.016$), so the economic reading above is robust to the objective. The two accepted sets agree on 1,579 days and differ on 200, split 134 price-only and 66 IV-only, all near the pegging boundary.

The lesson is that the objective is not the binding choice. Aligning the optimiser with the acceptance metric does not relieve the pegging and slightly aggravates it. Pegging is a structural degeneracy in the (κ, ρ, η) skew subspace, and it survives a change of residual. The IV-space objective weighs the deep-OTM wings more heavily than the price objective, since it no longer discounts them by their small prices. It therefore asks for more skew and reaches the box wall more often. Feller violation rises from 99.3% to 99.7% of accepted days and the median η edges up, both consistent with that pressure. One cost is specific to abandoning the price-space objective. The returned relative-price error is no longer controlled, and on a few accepted days it is large (maximum $J_{px} = 36.3$ against 0.32 under the price objective, with 18 days above 0.2 against 2). Those days are tight in vol points yet far off in relative price on a near-worthless wing cell. The practical reading is that the objective knob does not move acceptance. The levers that would are the identification fixes: a higher maturity floor, liquidity weighting, and anchoring κ .

10. Example option prices

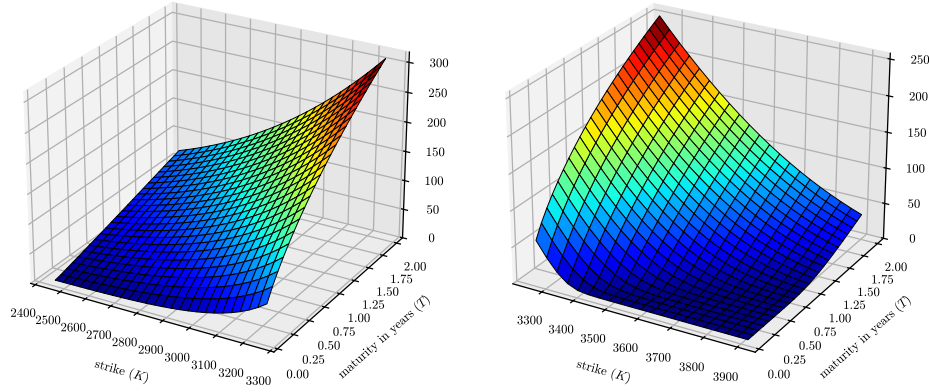


FIGURE 1. Heston OTM option prices for S_{ref} 3254.0554 on March 16, 2020: puts wing (left) and calls wing (right).

References

- Black, F. (1976). The Pricing of Commodity Contracts. *Journal of Financial Economics*, 3(1–2), 167–179. [https://doi.org/10.1016/0304-405X\(76\)90024-6](https://doi.org/10.1016/0304-405X(76)90024-6).
- Black, F., and Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637–654. <https://doi.org/10.1086/260062>.
- Feller, W. (1951). Two Singular Diffusion Problems. *Annals of Mathematics*, 54(1), 173–182. <https://doi.org/10.2307/1969318>.
- Heston, S. L. (1993). A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options. *The Review of Financial Studies*, 6(2), 327–343. <https://doi.org/10.1093/rfs/6.2.327>.
- Marquardt, D. W. (1963). An Algorithm for Least-Squares Estimation of Nonlinear Parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2), 431–441. <https://doi.org/10.1137/0111030>.
- Mikhailov, S., and Nögel, U. (2003). Heston’s Stochastic Volatility Model: Implementation, Calibration and Some Extensions. *Wilmott Magazine*, July 2003, 74–79.
- Ballabio, L., and the QuantLib contributors. *QuantLib: a free/open-source library for quantitative finance*, version 1.35. <https://www.quantlib.org>.

Appendix A. Algorithm

Algorithm 1 states the per-day routine; the driver applies it to every trading day and accumulates accepted rows into a single file (§7).

Algorithm 1 Per-day Heston calibration

Require: day d of OTM trades; rates $r(d), g(d)$; constants of Table 1

- 1: $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$ (pre-filter (13))
- 2: **if** $r(d)$ or $g(d)$ undefined on/before d **then**
- 3: **return** $\emptyset$ (no rate)
- 4: **end if**
- 5: $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$;
 $R \leftarrow \max_i S_i / \min_i S_i - 1$;
 $\text{HIGH_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6: $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$ with $m_i = K_i / S_i$ (re-strike (16))
- 7: $\mathcal{T} \leftarrow$ top- N_t maturities by volume; build $\mathcal{K}_\tau$ (n_k per wing);
 pivot to $\{\hat{\sigma}_{k,\tau}\}$, cells $\mathcal{N}$
- 8: **if** coverage (21) fails **then**
- 9: clear stale diagnostics; **return** $\emptyset$
- 10: **end if**
- 11: $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$; build seeds $\mathcal{S}$ (28)
- 12: **for** $s \in \mathcal{S}$ **do**
- 13: $\hat{\Phi}_s \leftarrow$ LM minimiser of $\sum_n (\varepsilon_n^{\text{px}})^2$ on $\mathcal{B}$ from s (29)
- 14: compute $J_{\text{iv}}(\hat{\Phi}_s)$ (26)
- 15: **end for**
- 16: $\Phi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Phi}_s)$ (select (30))
- 17: **if** $J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}}$ **and** $\neg \text{PEGGED}(\Phi^*)$ **then**
- 18: write repriced diagnostics; **return** $\text{row}(d, S_{\text{ref}}, r, g, \Phi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20: clear stale diagnostics; **return** $\emptyset$ (rejected)
- 21: **end if**
